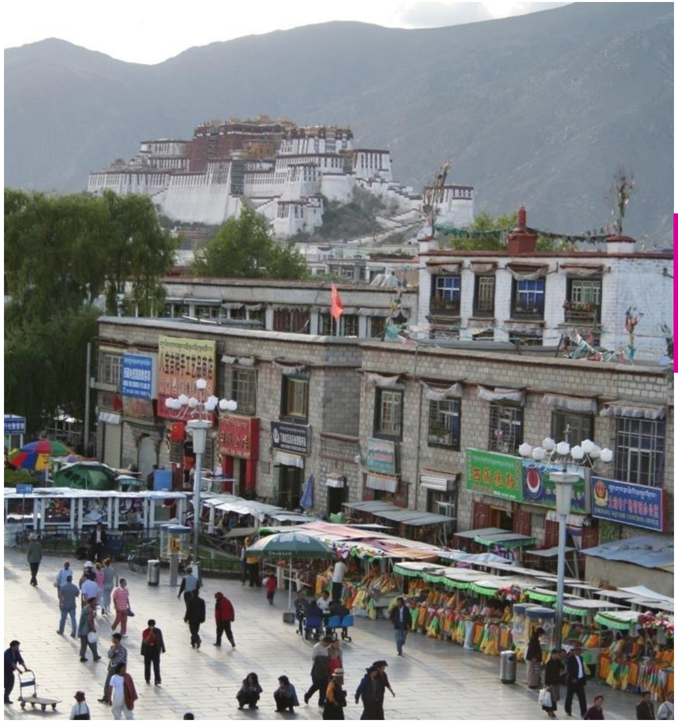
Today's Outline

- Multimodal Computer Vision: Image Recognition, Segmentation, and Detection

Specific recognition tasks

RGB Image.

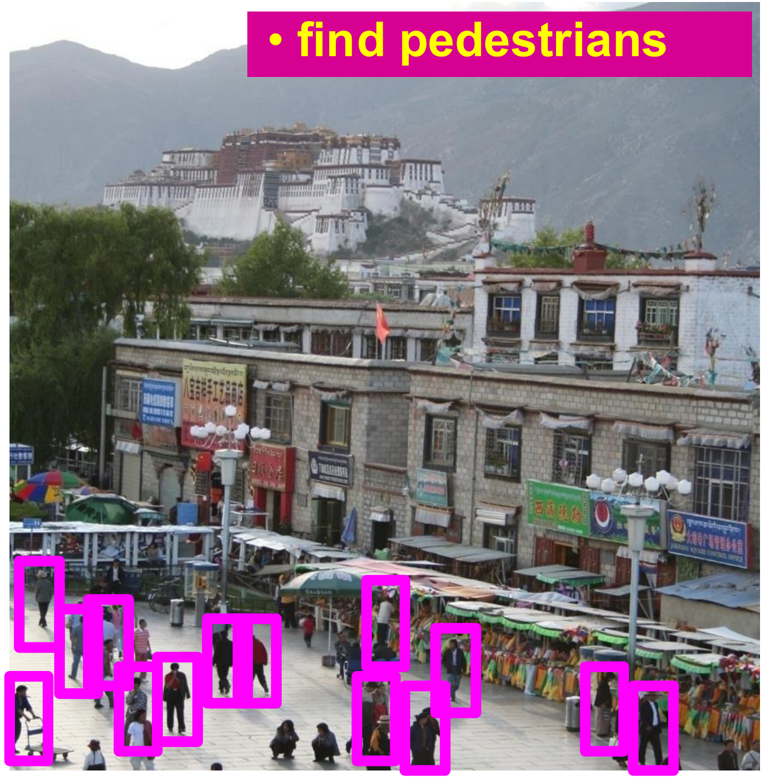
Scene categorization or classification



RGB Image.

- outdoor/indoor
- city/forest/factory/etc.

Object detection



Instance-wise annotation

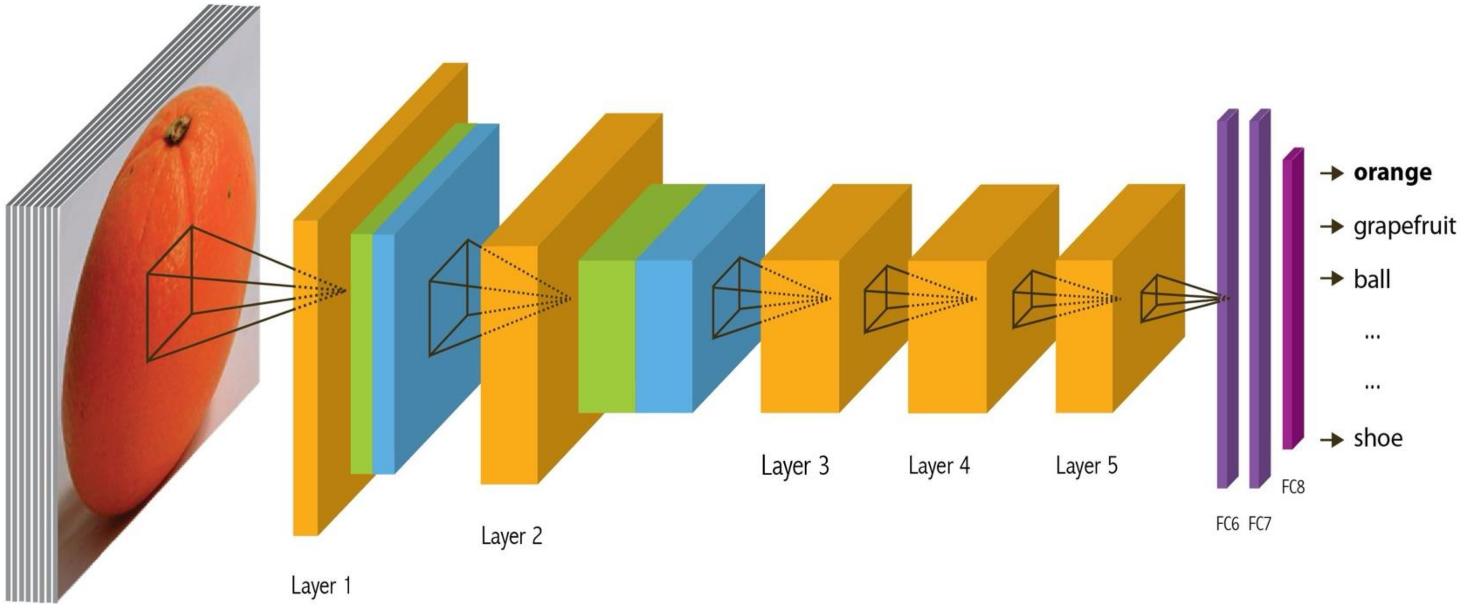
Image parsing / semantic segmentation

Pixel-wise annotation

Within-class variations

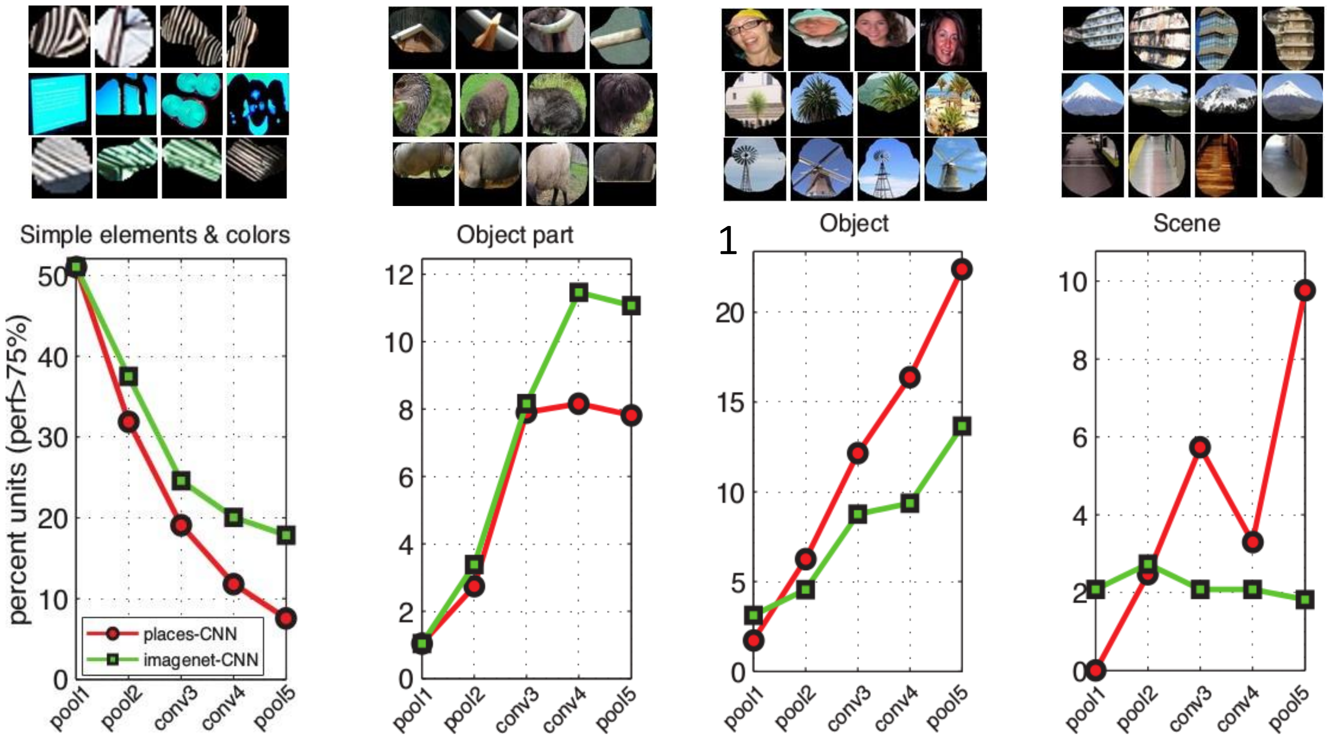
Classification problem.

Convolutional Network, AlexNet



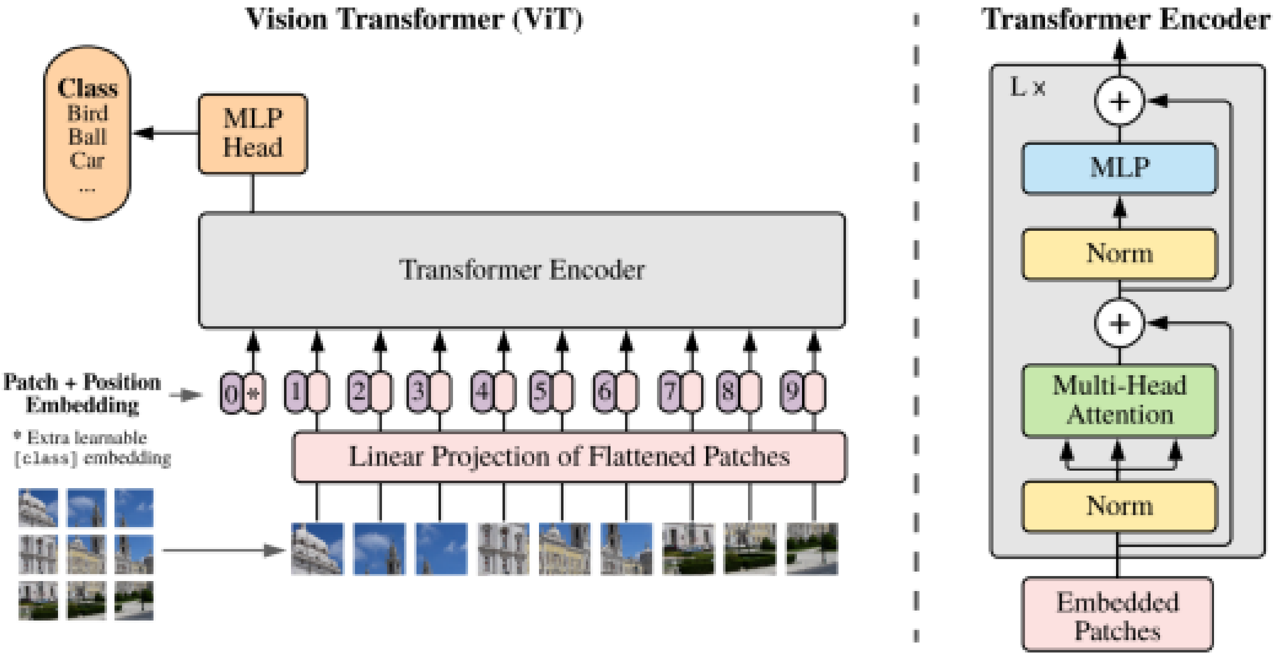
Model architecture.

Convolutional Network, AlexNet



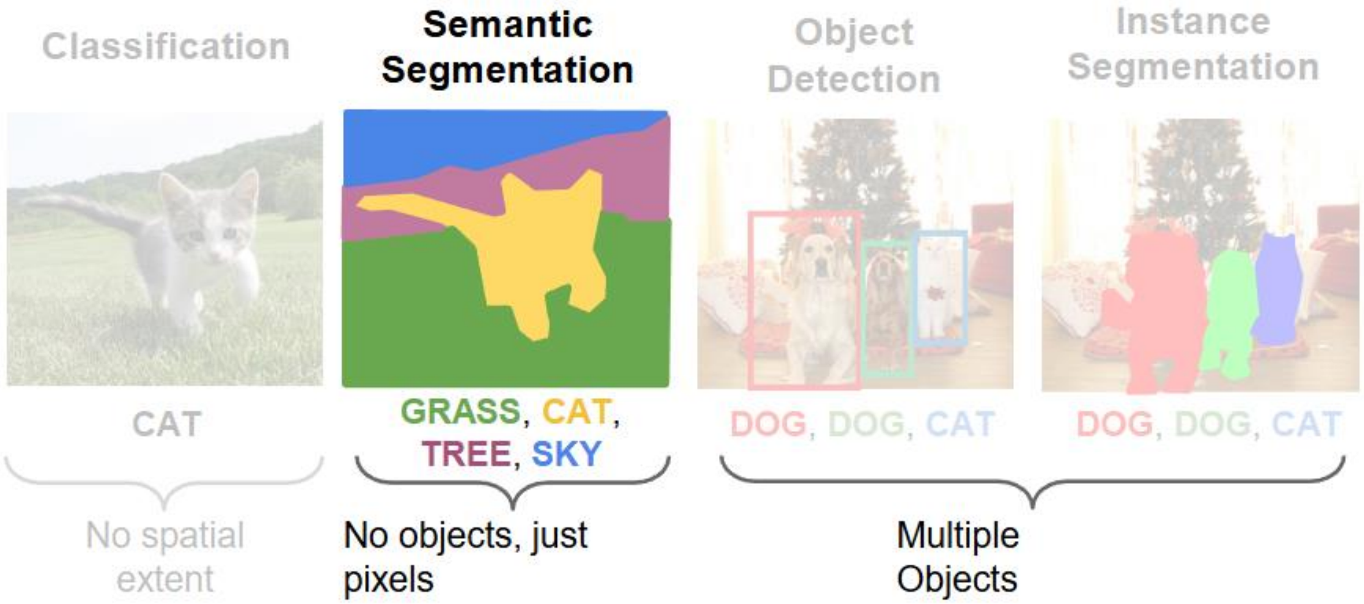
Object detectors emerge within CNN trained to classify scenes,
without any object supervision!

Transformer



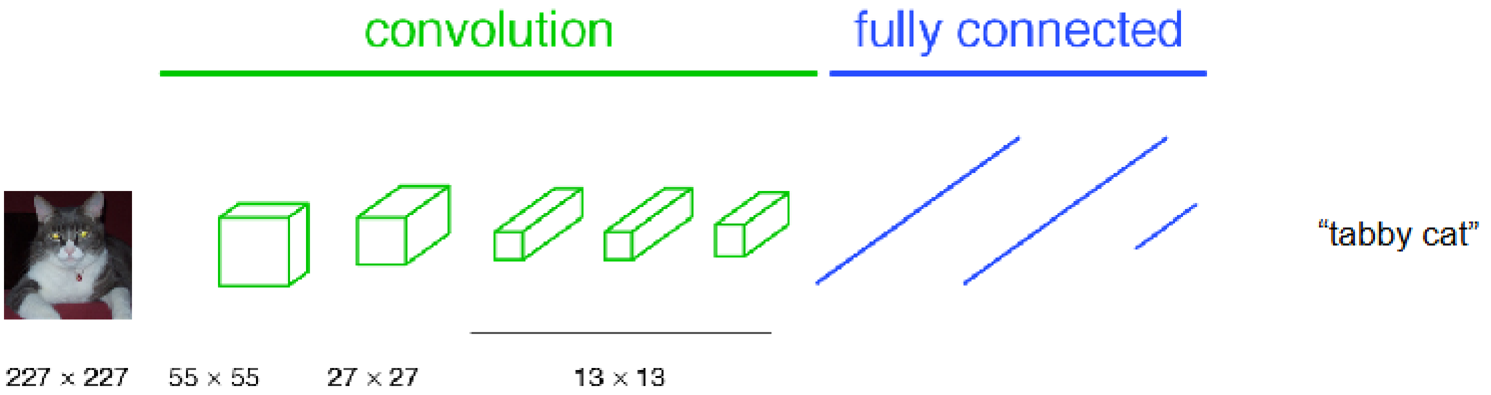
Classification using Transformer architecture.

Semantic Segmentation



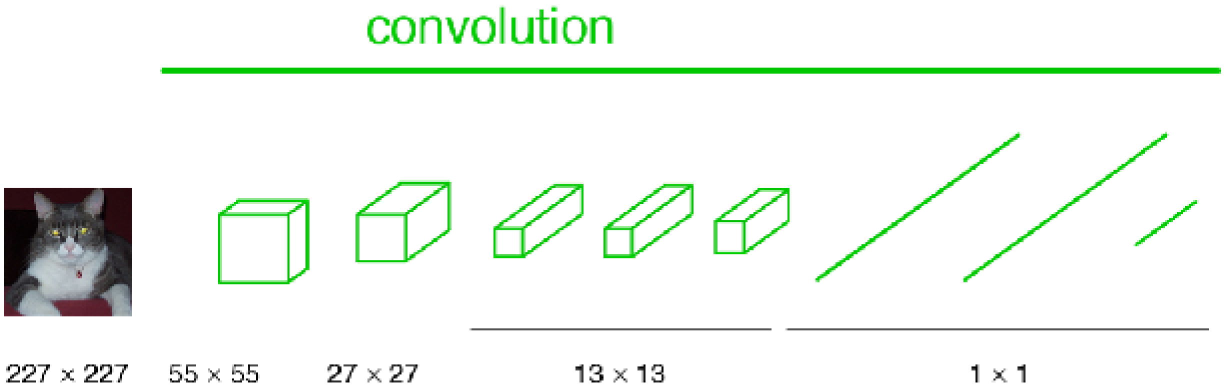
Semantic segmentation, the definition and difference.

Classification network



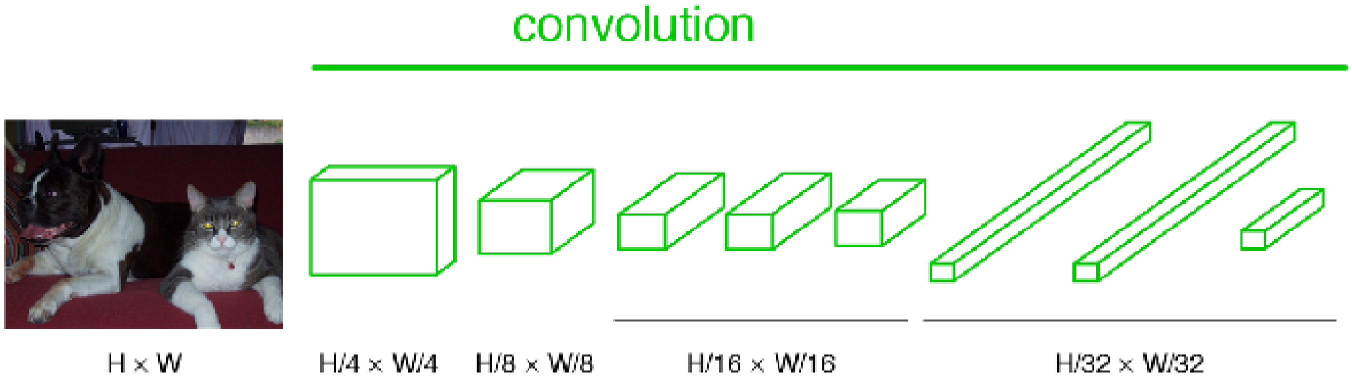
Typical classification network -> Segmentation network.

Fully convolutional network



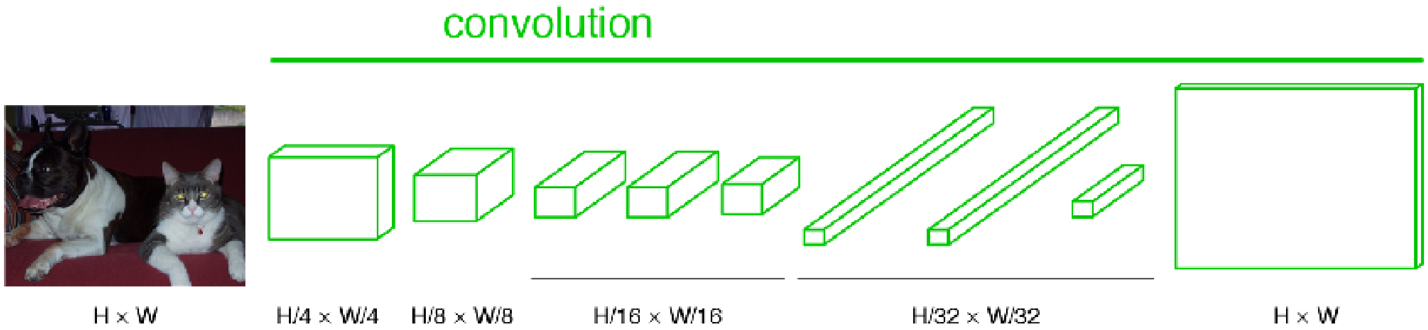
Note: “Fully Convolutional” and “Fully Connected” aren’t the same thing.They’re almost opposites, in fact.

Becoming fully convolutional



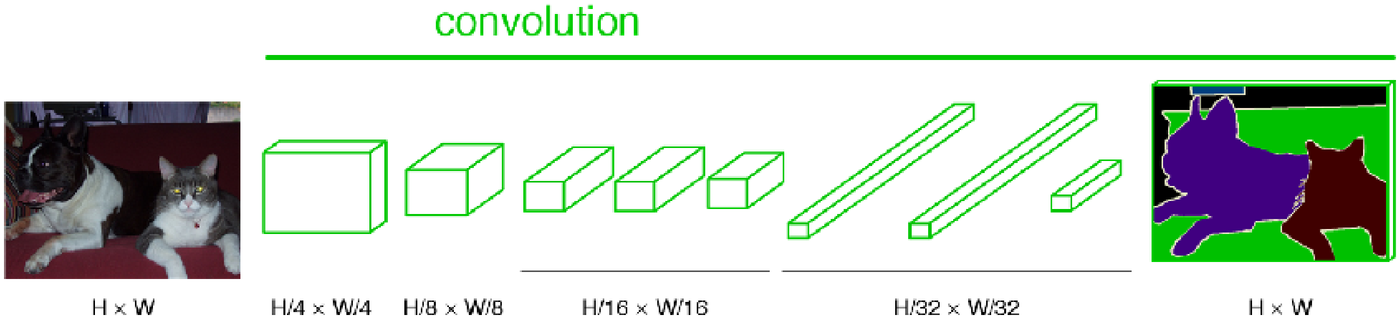
Typical classification network -> Segmentation network.

Upsampling output



Typical classification network -> Segmentation network.

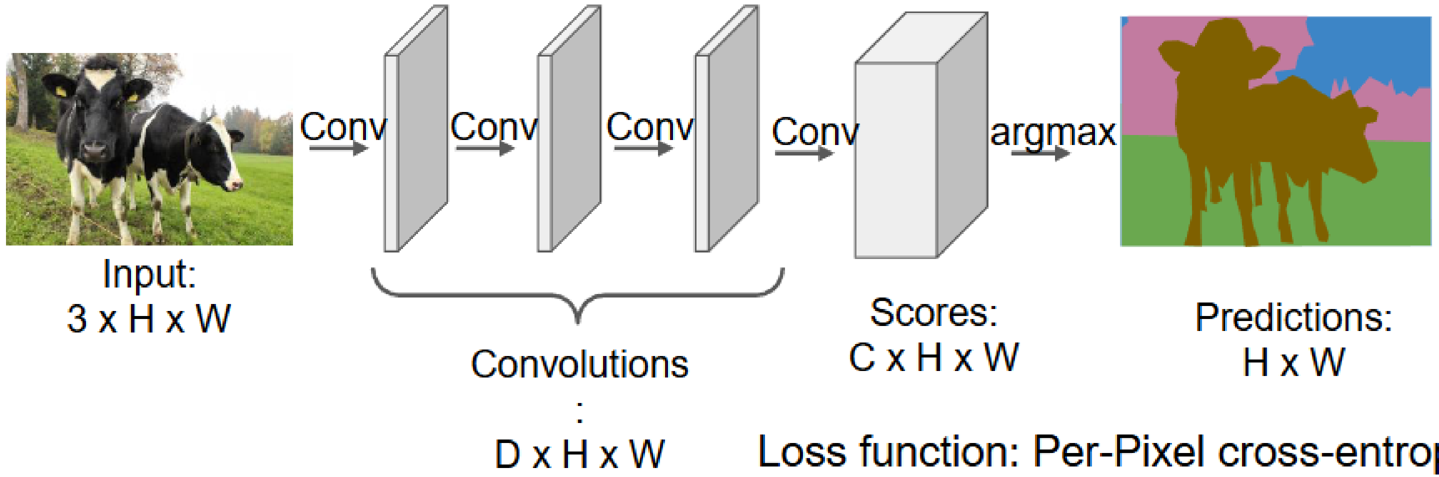
End-to-end, pixels-to-pixels network



Typical classification network -> Segmentation network.

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!

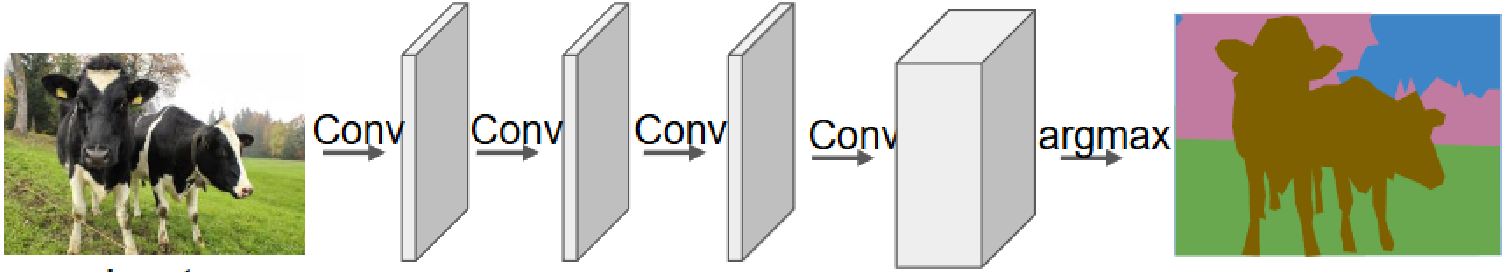


Loss function: Per-Pixel cross-entropy

Long et al, “Fully convolutional networks for semantic segmentation”, CVPR 2015

Fully Convolutional Network

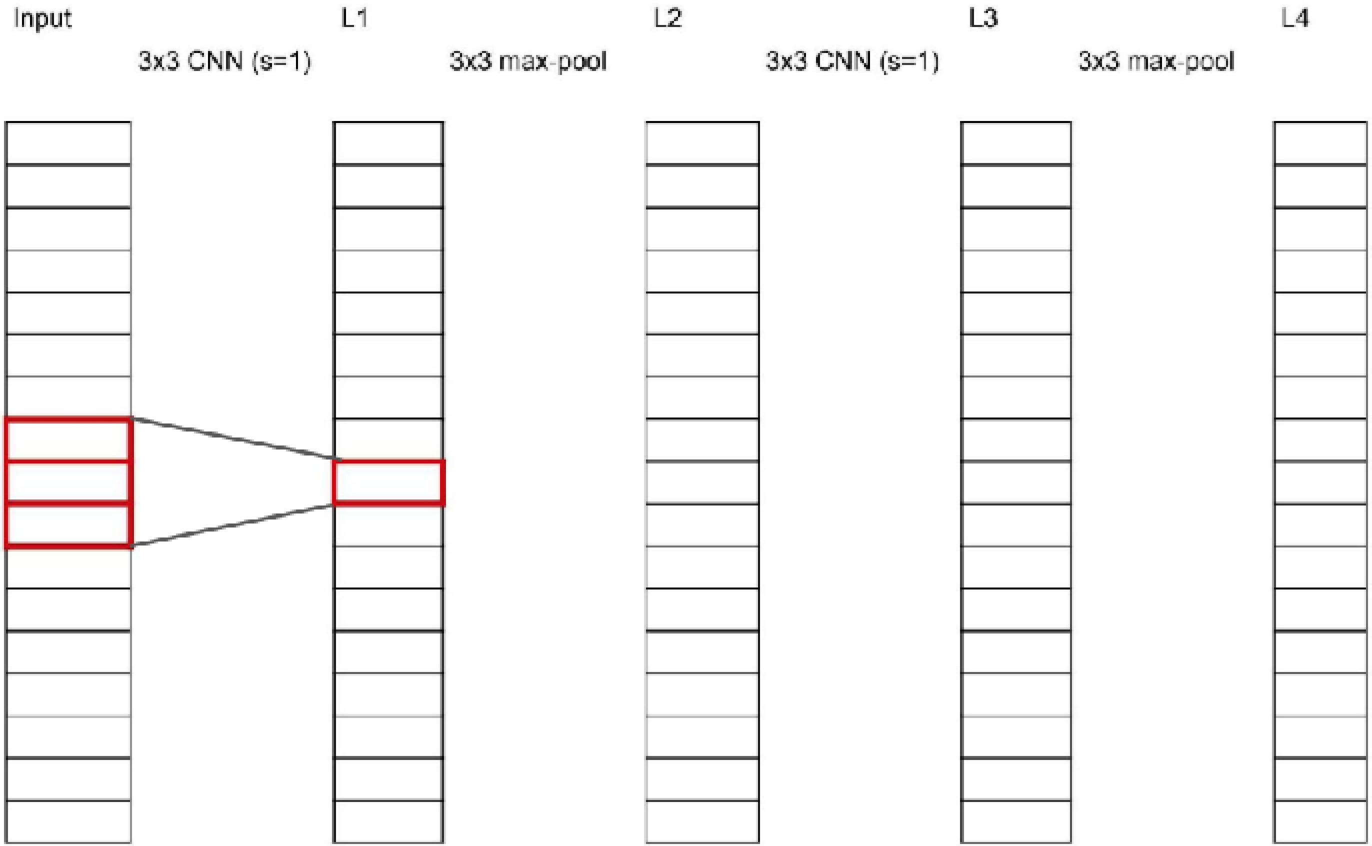
Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



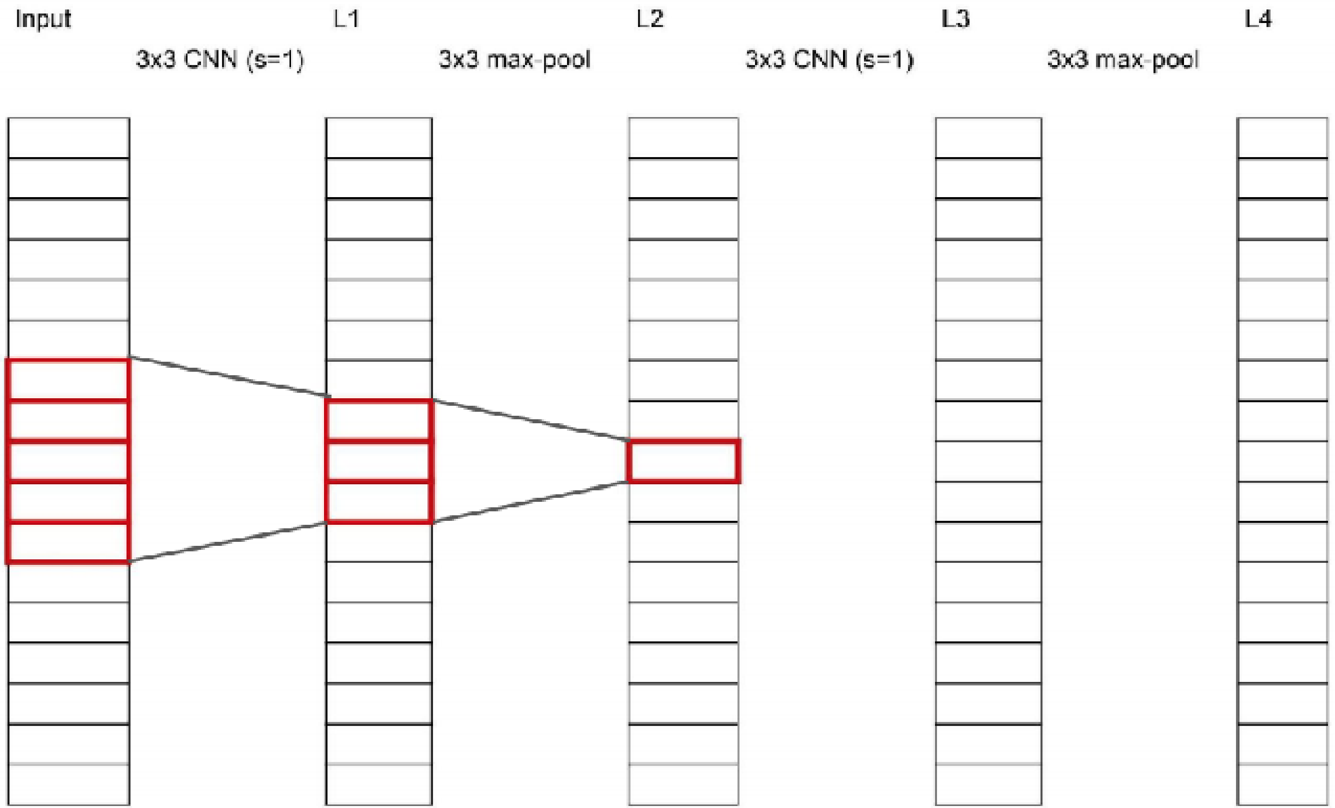
Input:
3 x H x W

Problem #1: Effective receptive field size is linear in number of conv layers: With L 3x3 conv layers, receptive field is 1+2L

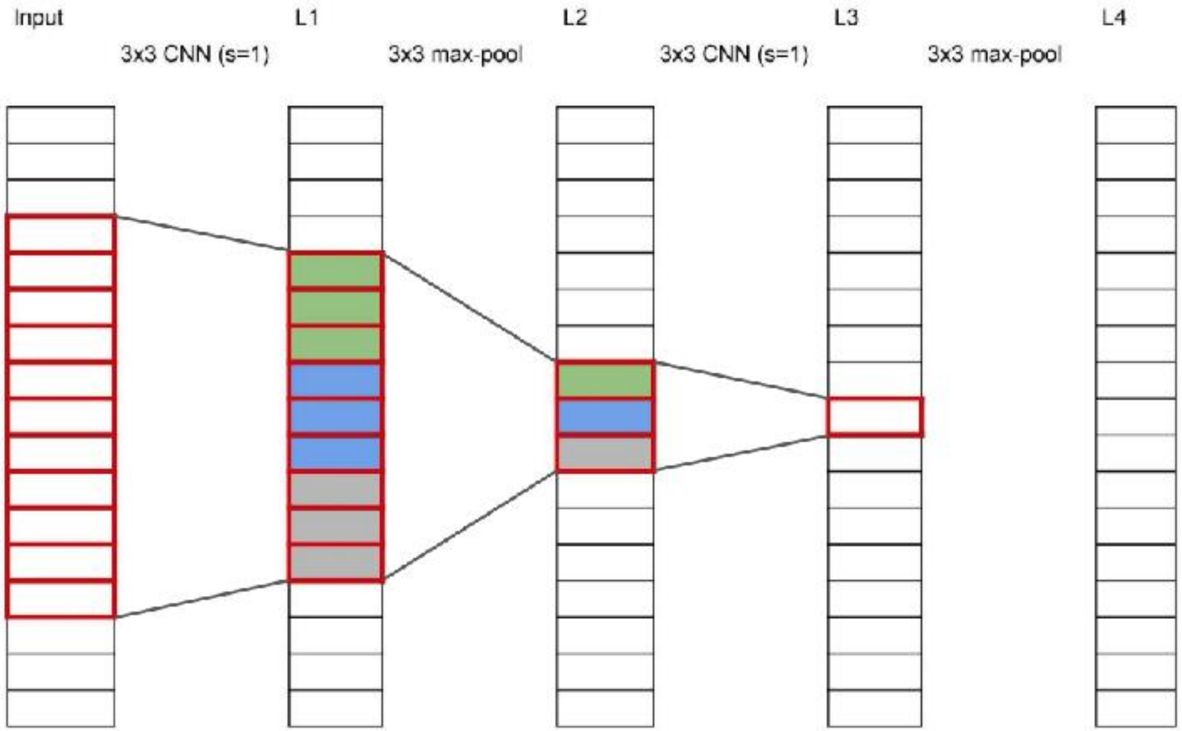
Receptive field



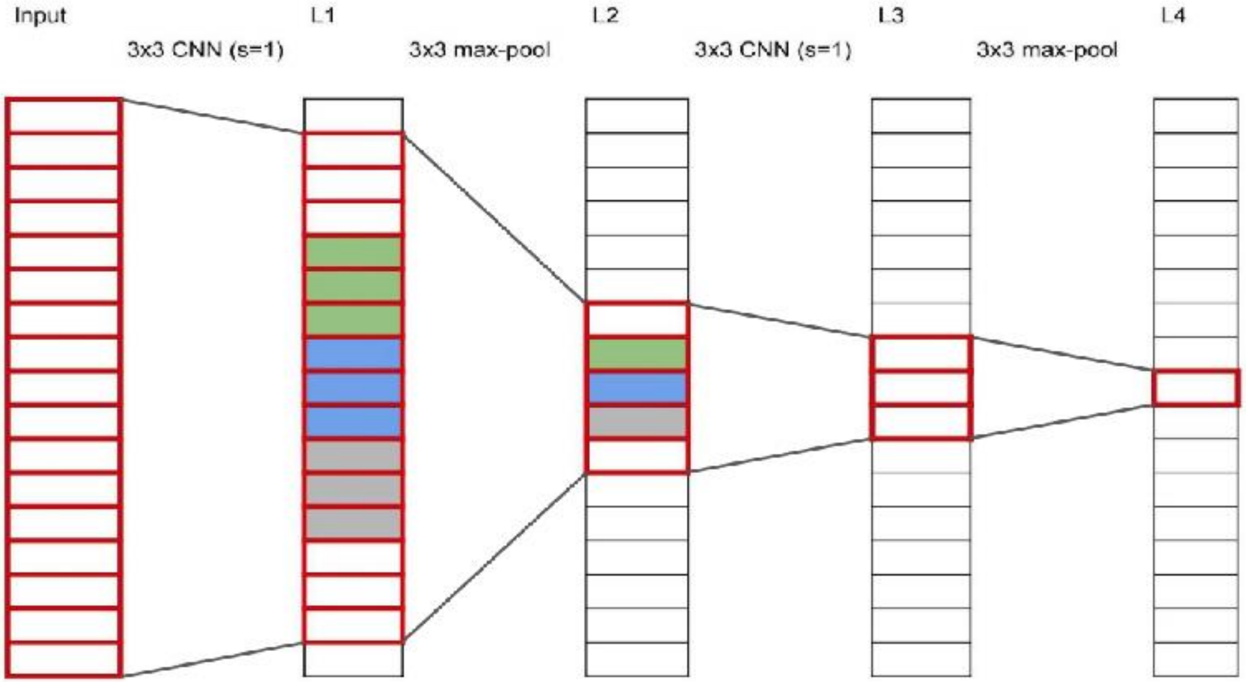
Receptive field



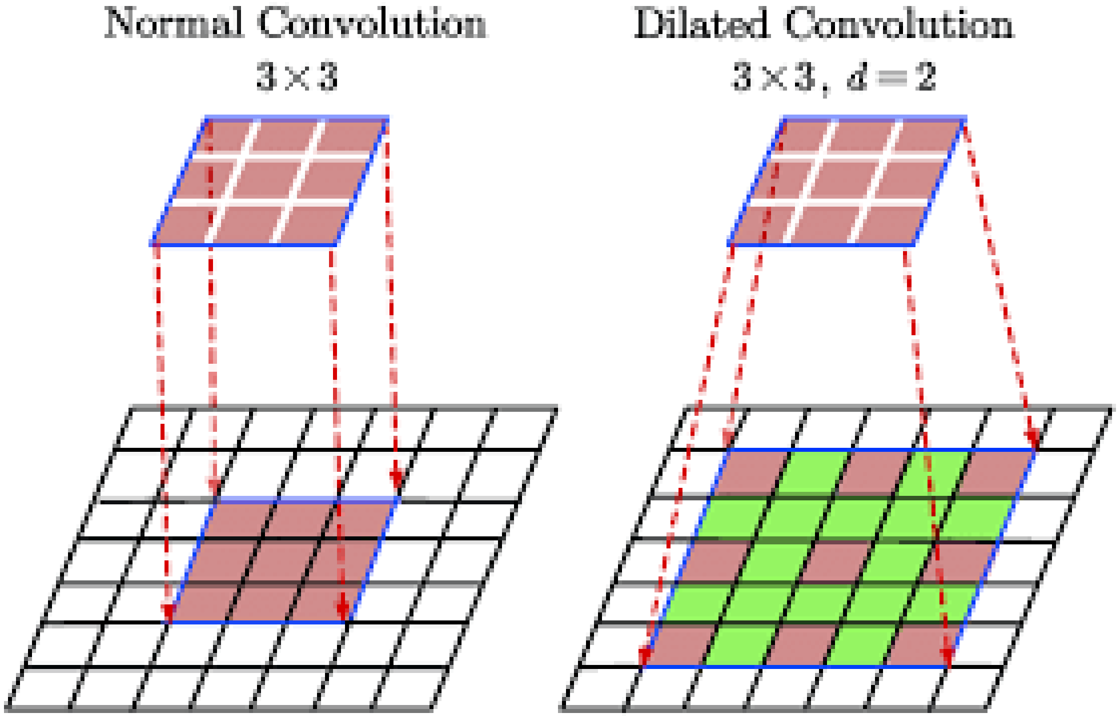
Receptive field



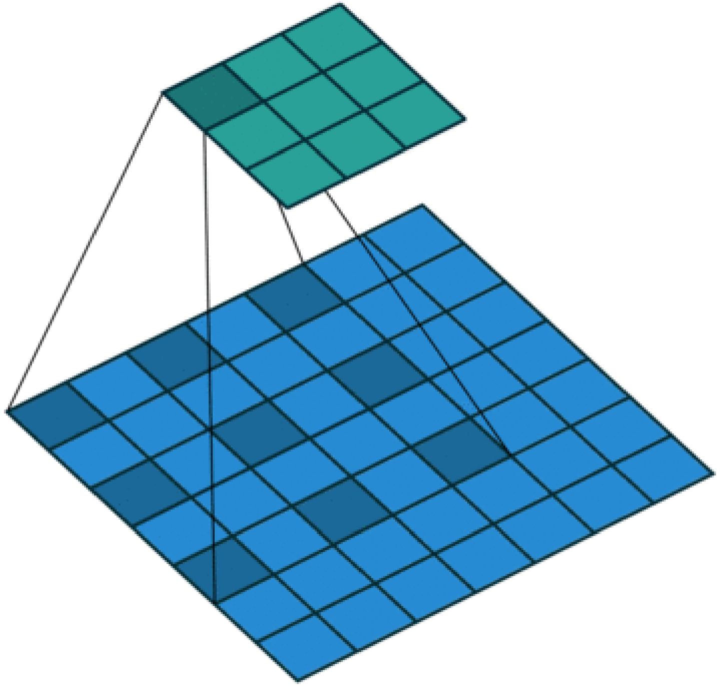
Receptive field



Dilated Convolution

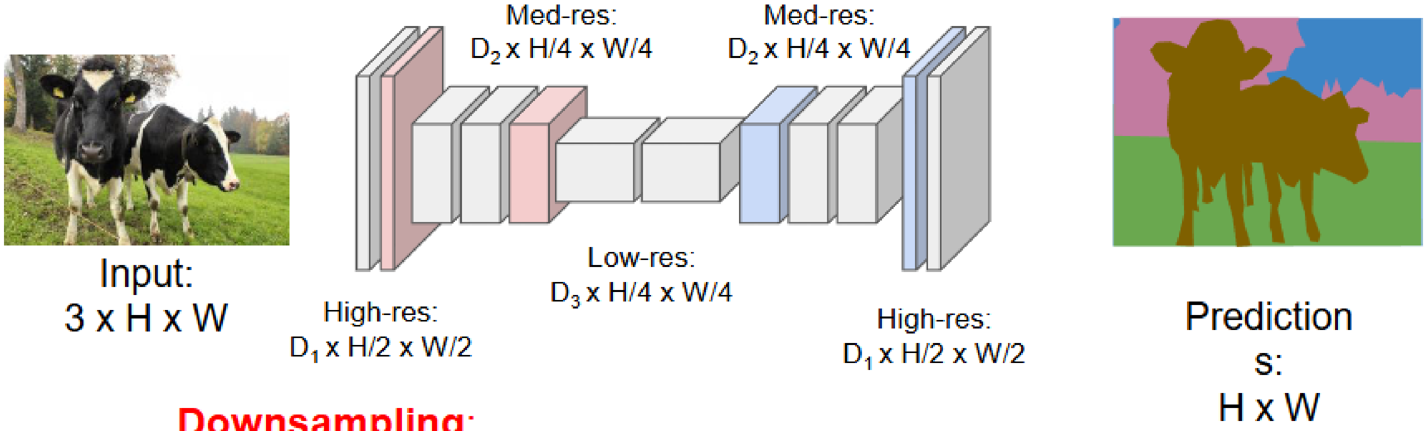


Dilated Convolution



Fully Convolutional Network

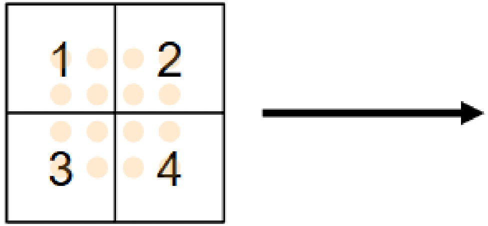
Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Downsampling:
Pooling, strided convolution

Upsampling:
???

Upsampling: Bilinear Interpolation



Input: C x 2 x 2

1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

Output: C x 4 x 4

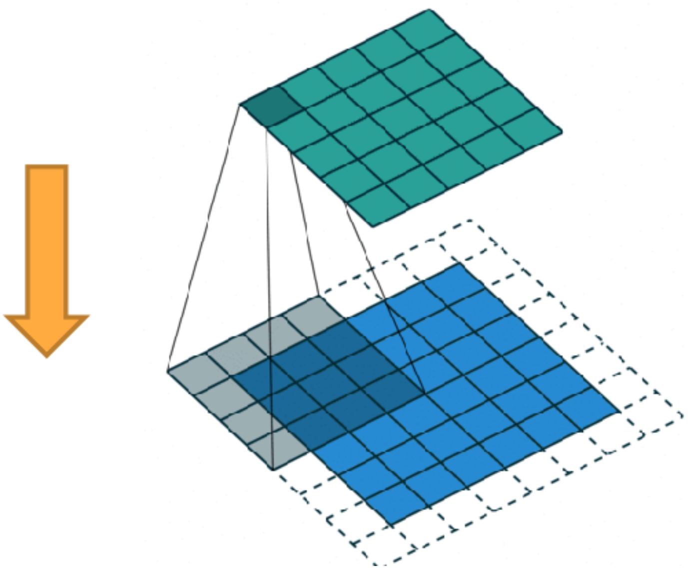
$$f_{x,y} = \sum_{i,j} f_{i,j} \max(0, 1 - |x - i|) \max(0, 1 - |y - j|) \quad \begin{array}{l} i \in \{\lfloor x \rfloor - 1, \ldots, \lceil x \rceil + 1\} \\ j \in \{\lfloor y \rfloor - 1, \ldots, \lceil y \rceil + 1\} \end{array}$$

Use two closest neighbors in x and y to construct linear approximations

Upsampling: Transpose Convolution

Sometimes called "Deconvolution";

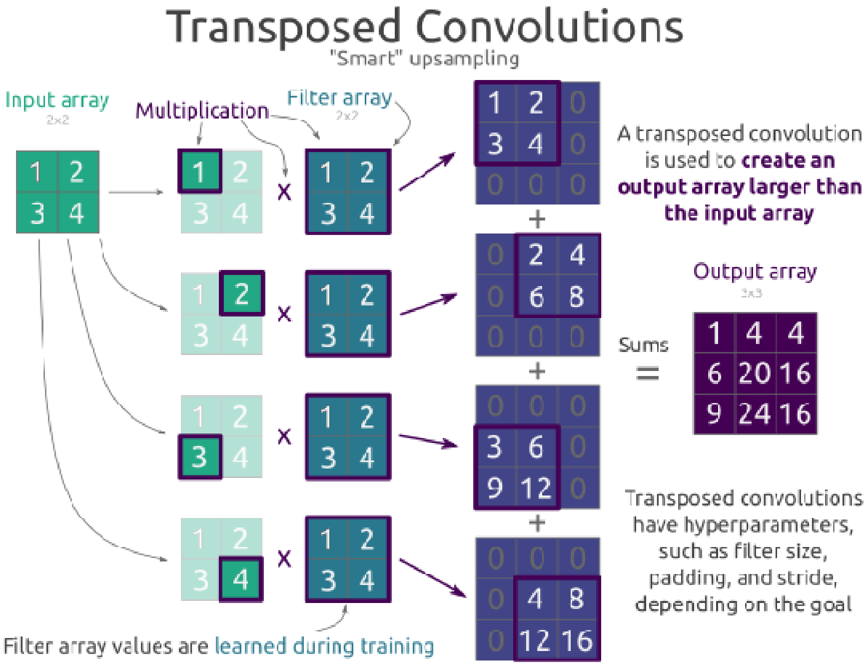
This case, the filter is 4x4 and the outer boundary of the output is unused



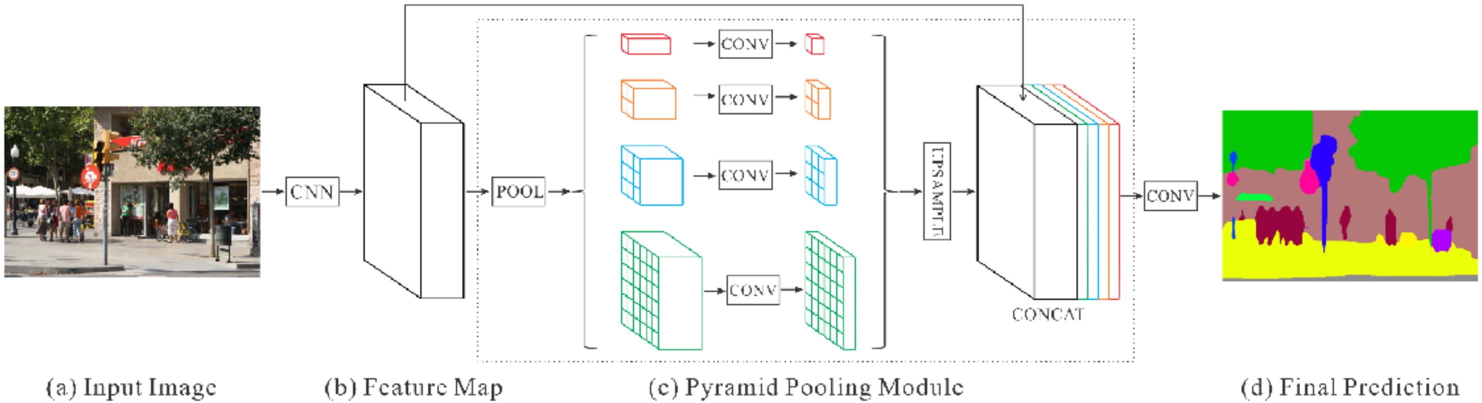
Upsampling: Transpose Convolution

Sometimes called "Deconvolution";

This case, the filter is 2x2 and the outer boundary of the output is unused



Example: Pyramid Scene Parsing Network



Framework overview of PSPNet

Example: Pyramid Scene Parsing Network

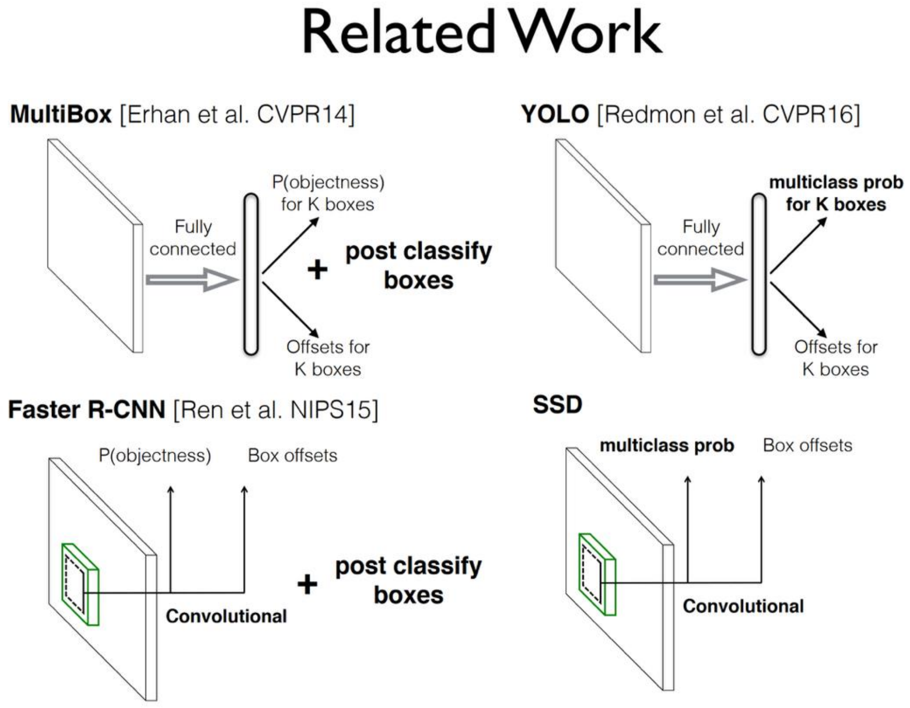
Segmentation example.

What if we want other types of outputs?

- An arbitrary number of bounding boxes (4 parameters each) or segmentation masks (hundreds of parameters each)
- we will examine influential methods for object detection

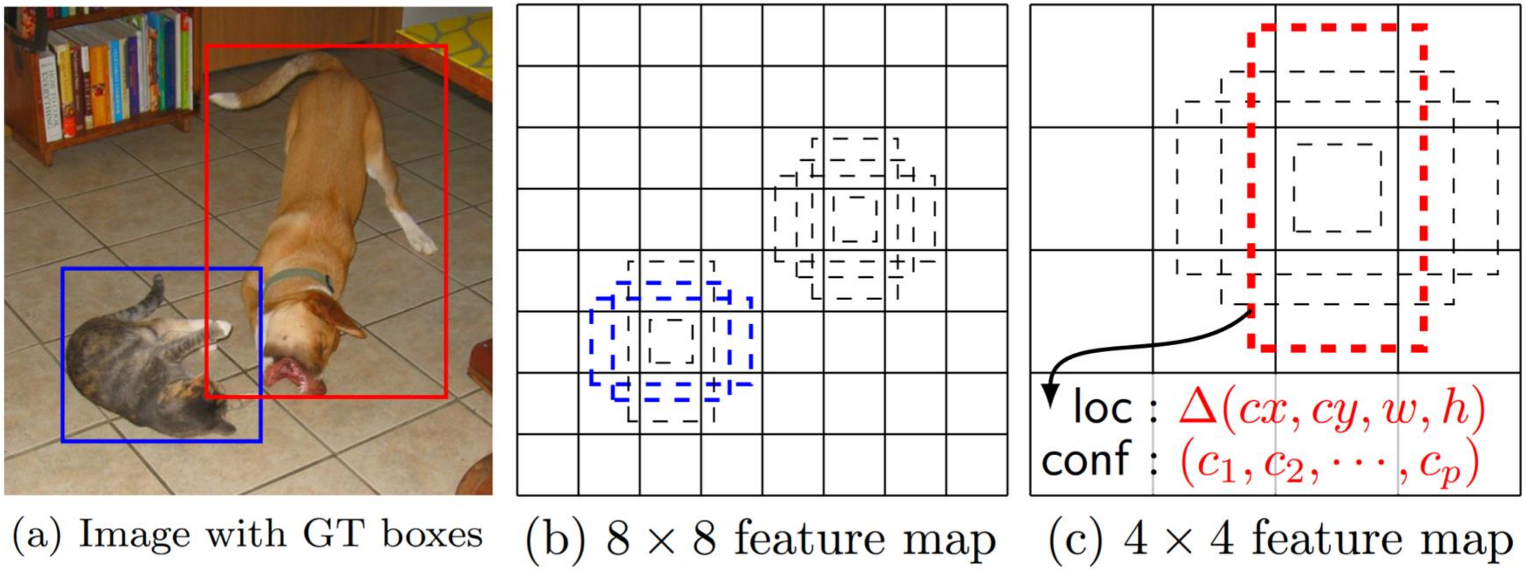
Segmentation example.

Object detection



Object detection principles.

Object detection

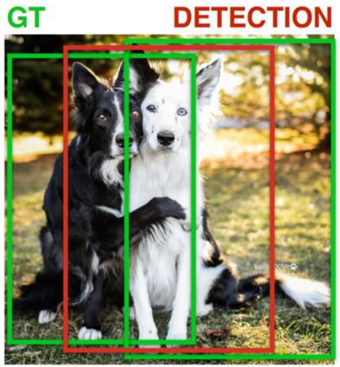


Object detection principles.

Object detection

Why So Many Default Boxes?

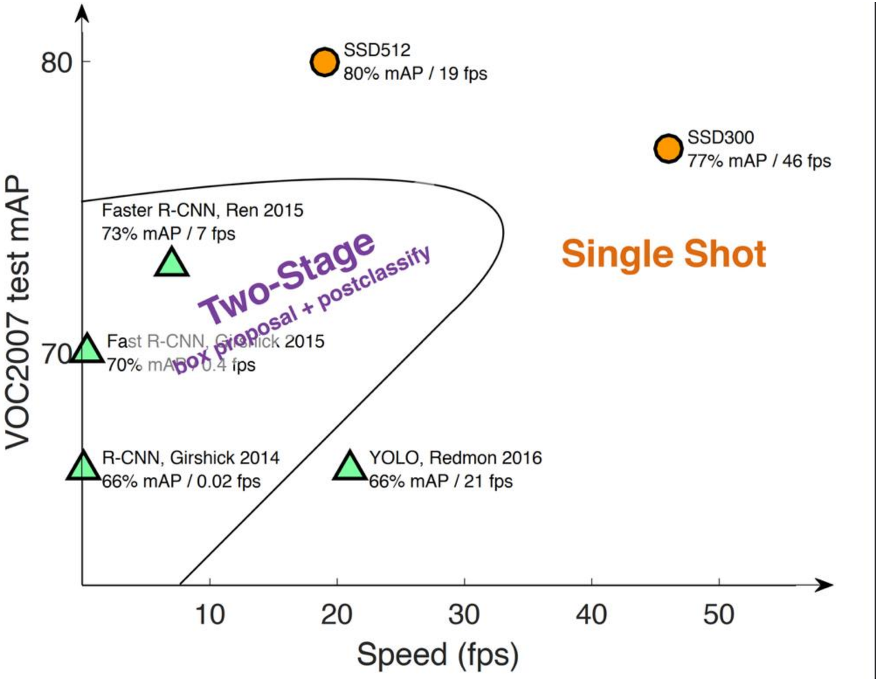
	Faster R-CNN	YOLO	SSD300	SSD512
# Default Boxes	6000	98	8732	24564
Resolution	1000x600	448x448	300x300	512x512



- SmoothL1 or L2 loss for box shape averages among likely hypotheses
- Need to have enough default boxes (discrete bins) to do accurate regression in each
- General principle for regressing complex continuous outputs with deep nets

Object detection principles.

Object detection

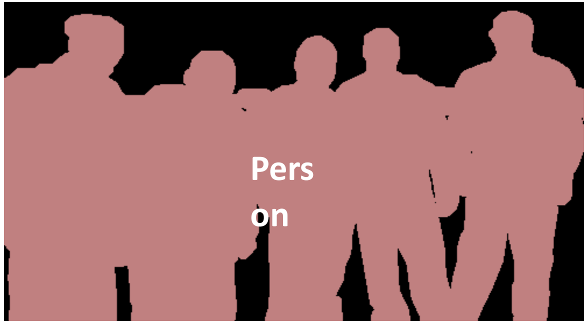


Object detection principles.

Mask R-CNN (ICCV 2017)

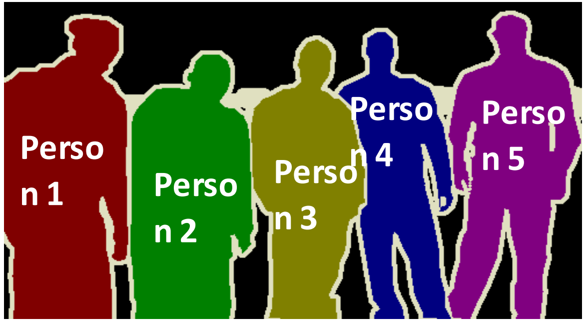
Object Detection

✓



Semantic Segmentation

✓

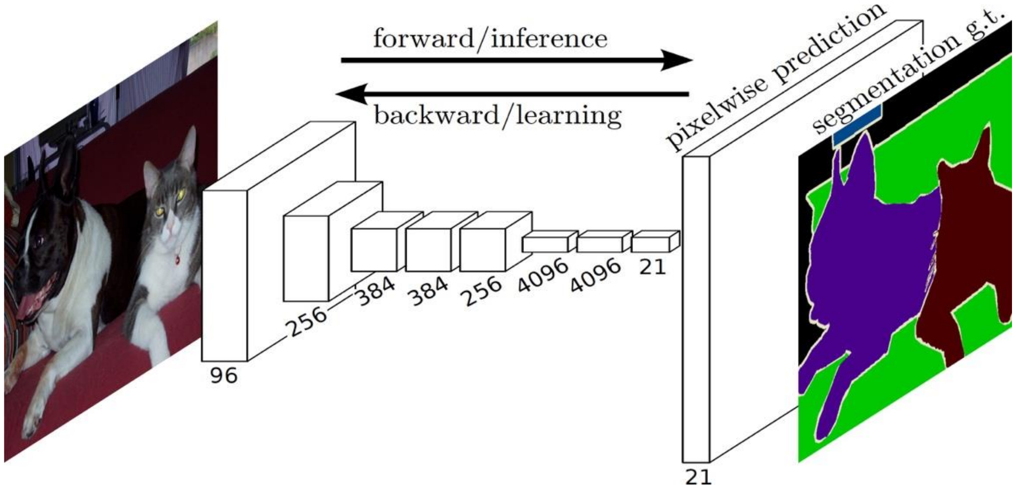


Instance Segmentation

?

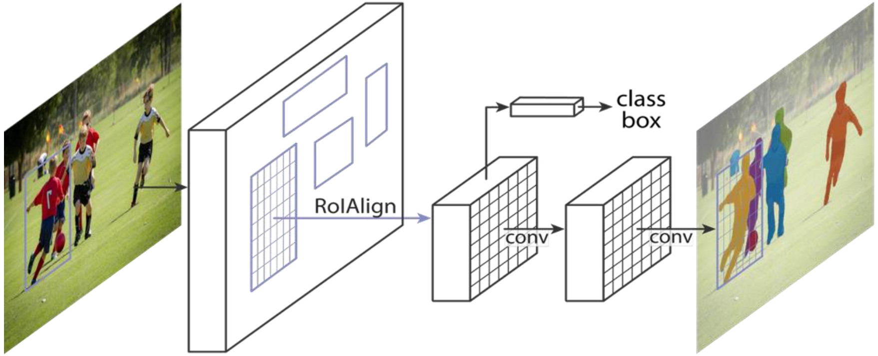
Goal of Semantic Segmentation

- Fully Convolutional Net (FCN)
  - ✓ Good speed
  - ✓ Good accuracy
  - ✓ Intuitive
  - ✓ Easy to use

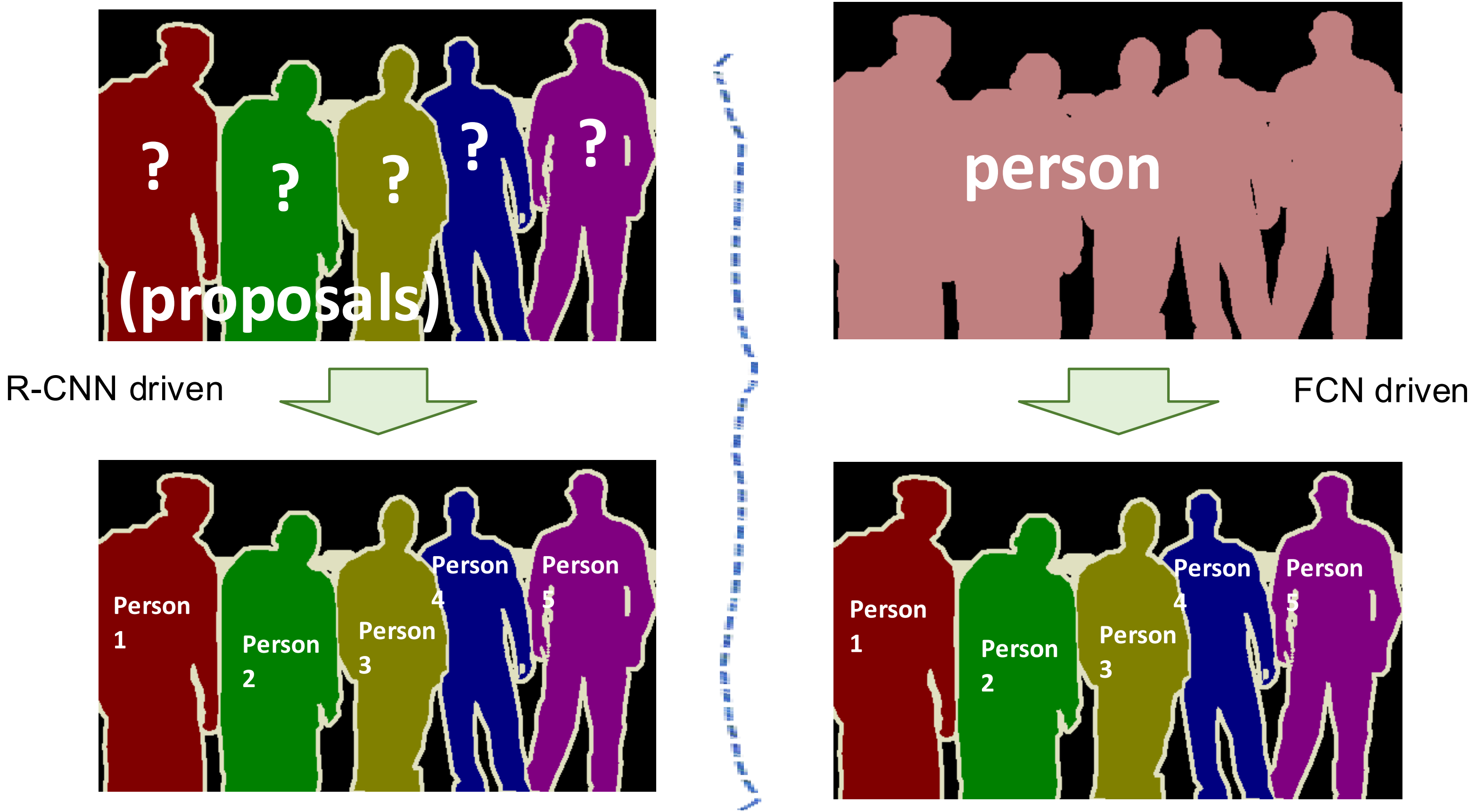


Goal of Instance Segmentation

- **Goals** of Mask R-CNN
  - ✓ Good speed
  - ✓ Good accuracy
  - ✓ Intuitive
  - ✓ Easy to use

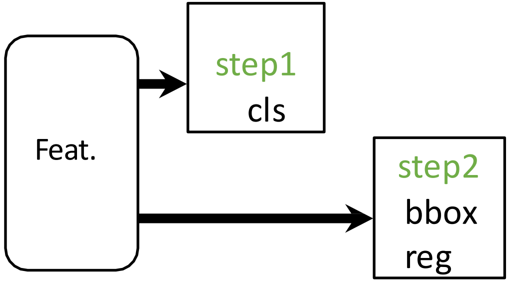


Instance Segmentation Methods

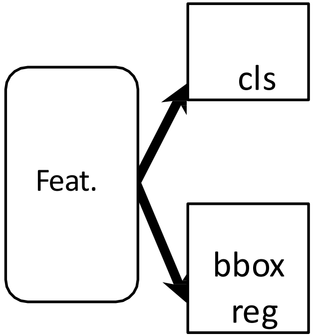


Parallel Heads

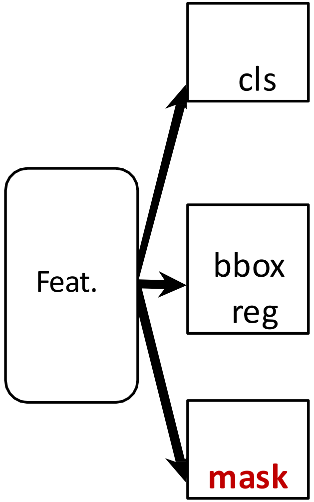
- Easy, fast to implement and train



(slow) R-CNN



Fast/er R-CNN



Mask R-CNN

Instance Segmentation Methods

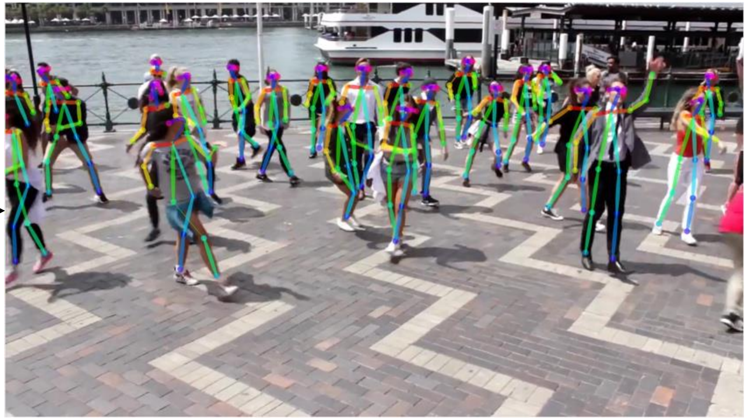
R-CNN driven

- SDS [Hariharan et al, ECCV'14]
- HyperCol [Hariharan et al, CVPR'15]
- CFM [Dai et al, CVPR'15]
- MNC [Dai et al, CVPR'16]

FCN driven

- PFN [Liang et al, arXiv'15]
- InstanceCut [Kirillov et al, CVPR'17
- Watershed [Bai & Urtasun, CVPR'17
- FCIS [Li et al, CVPR'17]
- DIN [Arnab & Torr, CVPR'17]

Human Pose Estimation

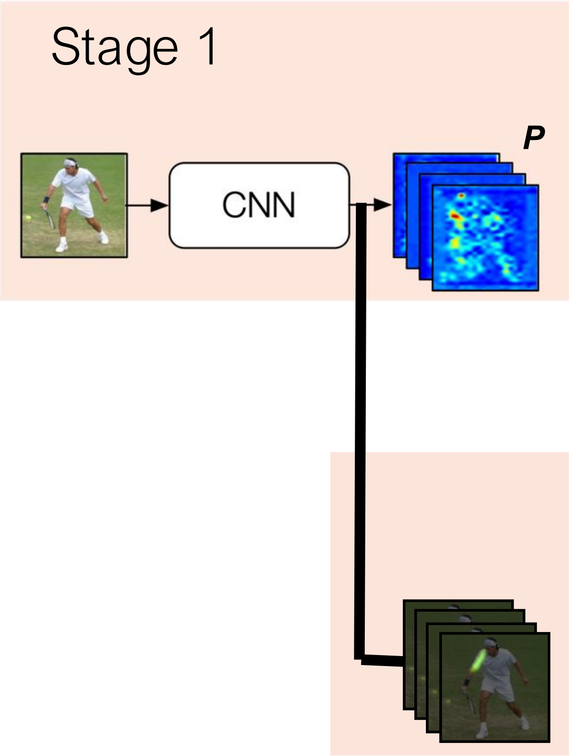


Human Pose Estimation

Challenges:

- Unknown number of people
- Variance in person scales
- Occlusion between people

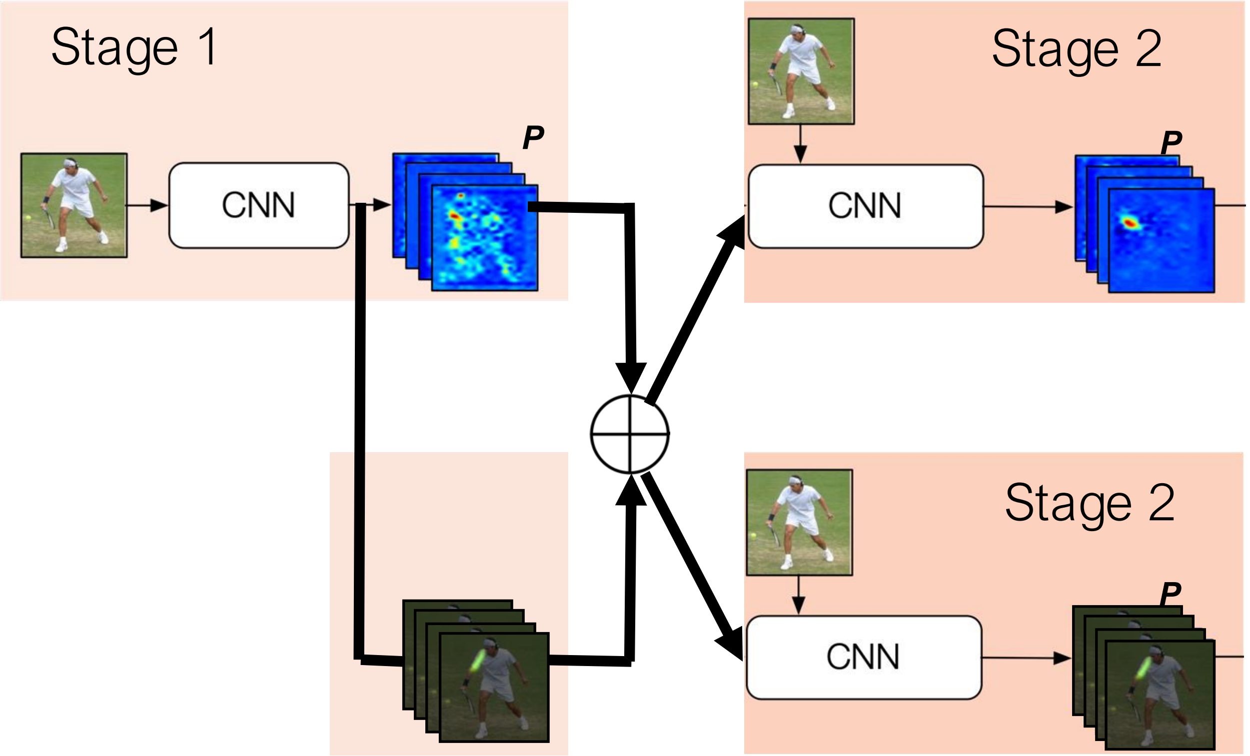
Jointly Learning Parts Detection and Parts Association



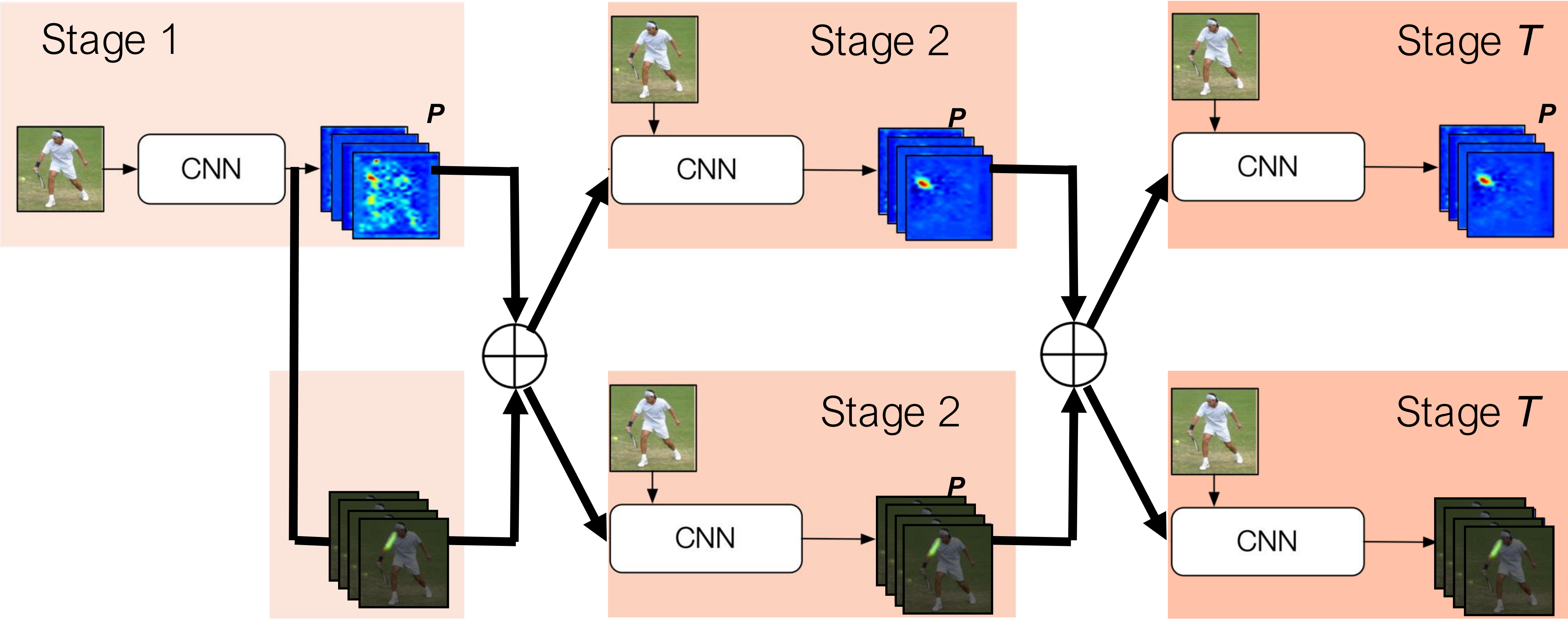
1st branch
part heatmaps

2nd branch
part affinity fields

Jointly Learning Parts Detection and Parts Association



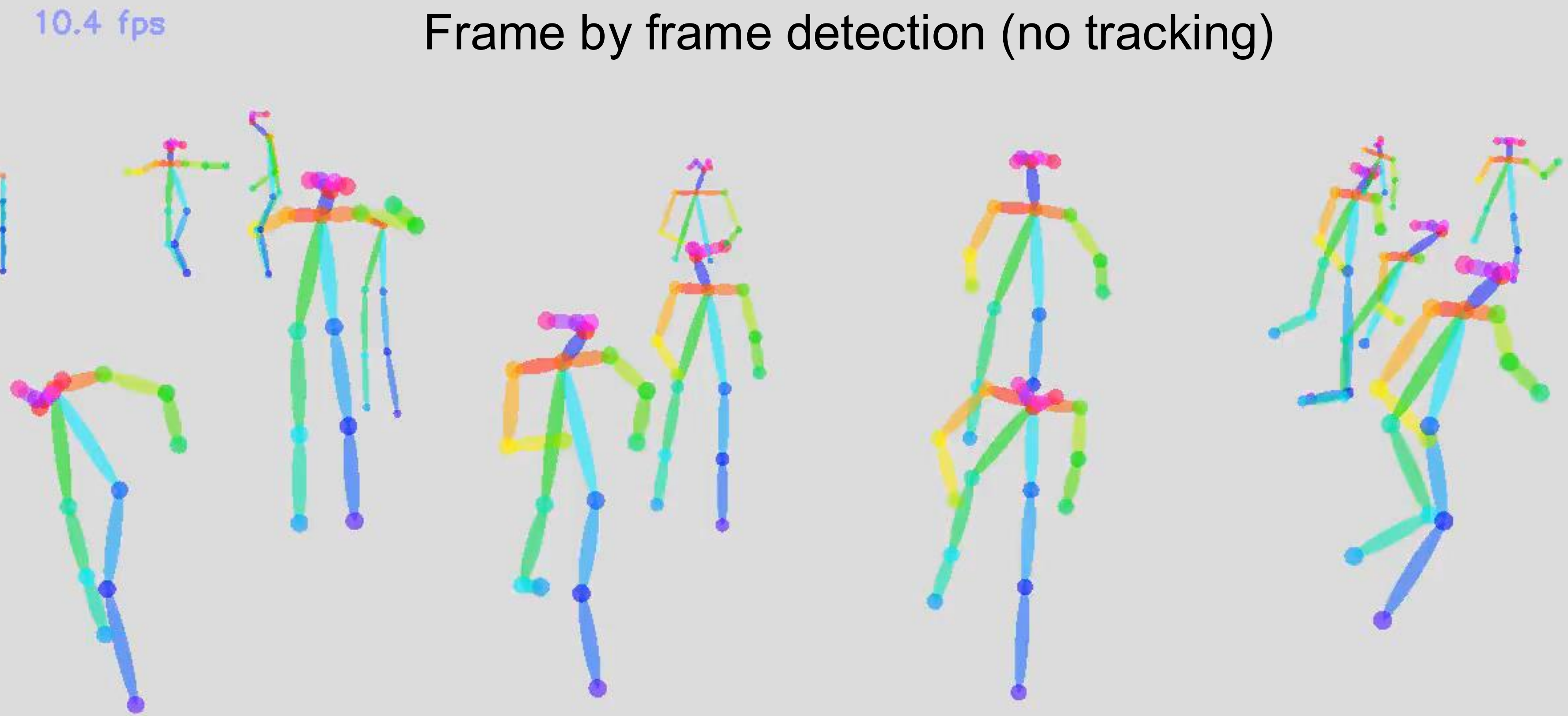
Jointly Learning Parts Detection and Parts Association




Bkg


PAFs

Frame by frame detection (no tracking)



Source: https://www.youtube.com/watch?v=2DiQUX11YaY

ECEN 689: Paper Presentation Logistics

Presentation Scope

- 1-2 papers on a chosen topic
- Duration: 30 minutes

Coverage

- Problem definition and motivation
- Existing approaches
- Proposed method
- Experimental results
- Limitations and future work

ECEN 689: Paper Presentation Logistics

Link:

https://docs.google.com/spreadsheets/d/1ay_U54jgY0d6U6-ftUmhsSSfginHFWd9p6pnvtf1fsk/edit?usp=sharing